

Introduction to Machine Learning

Fundamentals

- **Artificial intelligence** is the science of training machines to perform human tasks.
- Subsets of artificial intelligence (Hurwitz, 2018):
 - **Reasoning** - making inferences based on data
 - **Natural Language Processing** (NLP) – understanding both written text and human speech
 - **Planning** – acting autonomously and flexibly to construct a sequence of actions to reach a final goal
 - **Machine Learning**
- **Machine learning** is a form of artificial intelligence that enables a system to learn from data, identify patterns, and make decisions with minimal human intervention.
- Common examples of machine learning:
 - When you search and view products in an online shopping site, you may encounter other similar products that you may find interesting the next time you go online.
 - Virtual assistants like Siri and Alexa, are asked over voice when you need to find information.
 - Transportation apps suggest faster routes and computes prices of rides based on the rider demand.
 - Facebook suggests people that you may want to become friends with and recognizes you and your friends' faces when you upload photos.
 - Chatbots serve as customer service agents when you can't reach the company hotline.
- Purpose of machine learning:
 - **Classification** – classifying an item into a distinct set of categories (*Ex. Recognizing the object in an image*)
 - **Clustering** – grouping items with common properties (*Ex. Facebook suggesting people to add in a group*)
 - **Prediction** – forecasting future values (*Ex. Demand for products around holiday season*)
- A **machine learning model** is the output generated when you train your machine learning algorithm with data. (Hurwitz, 2018)

Machine Learning Models

- **Supervised Learning:** The machine is provided with an established set of data (labeled data). The categories of supervised learning are:
 - **Classification**
 - **Regression** – predicting the values of a continuous variable
Ex. Weather forecasting provides a prediction on the weather based on historical weather patterns and the current conditions.
- **Unsupervised learning:** The machine is provided with a set of data but is not provided with a specific answer (unlabeled data). The machine may identify trends of similarity through clustering.
Ex. Identifying if an e-mail is a spam or not
- **Semi-supervised learning:** The machine learns from a dataset that includes both labeled and unlabeled data, usually most unlabeled.
- **Reinforcement learning:** The machine learns through trial and error. It is often used for robotics, gaming, and navigation. It has three (3) primary components.
Ex. A self-driving car that picks the best lane to go to
 - **Agent** – the learner or decision maker (self-driving car)
 - **Environment** – everything the agent interacts with (streets other cars, human drivers, etc.)
 - **Action** – what the agent can do (drive, pick lane, etc.)

Sample Programs

• Java

```
import java.util.ArrayList;
import java.util.Collections;
import java.util.Scanner;

public class Chatbot {
    public static void main(String args[]) throws InterruptedException {
        Scanner input = new Scanner(System.in);
        System.out.print("What is your name? ");
        String name = input.nextLine();
        Thread.sleep(2000);
        System.out.println("Nice to meet you, " + name + ". I'm Nika.");
        System.out.print("How are you today? ");
        Thread.sleep(2000);
        String feeling = input.nextLine();
        if (feeling.equalsIgnoreCase("good"))
            System.out.println("Wow! I feel the same way.");
        else if (feeling.equalsIgnoreCase("great"))
            System.out.println("That's nice to know!");
        else
            System.out.println("Go ahead and do something you love.");
        Thread.sleep(2000);
        System.out.print("What is your favorite color? ");
        String fave = input.nextLine();
        ArrayList colors = new ArrayList();
        Collections.addAll(colors, "orange", "yellow", "purple");
        Collections.shuffle(colors);
        Thread.sleep(2000);
        System.out.println(fave + " is cool! Mine is " + colors.get(0) + ".");
    }
}
```

Sample Output:

```
Hello, what is your name? Mark
Nice to meet you, Mark. I'm Nika.
How are you today? Great
That's nice to know!
What is your favorite color? Blue
Blue is cool! Mine is yellow.
```

• Python

```
import time
import random

name = input("Hello, what is your name? ")
time.sleep(2)
print("Nice to meet you, " + name + ". I'm Nika.")

feeling = input("How are you today? ")
time.sleep(2)
if "good" in feeling.lower():
    print("Wow! I feel the same way.")
elif "great" in feeling.lower():
    print("That's nice to know!")
else:
    print("Go ahead and do something you love.")

time.sleep(2)
fave = input("What is your favorite color? ")
colors = ["orange", "yellow", "purple"]
time.sleep(2)
print(fave + " is cool! Mine is " + random.choice(colors) + ".")
```

References:

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